PRACTICAL 7. CREATING INTERACTIVE DASHBOARDS AND DATA STORIES IN TABLEAU

Aim:- To create an interactive dashboard and data story for analyzing retail store performance using Tableau.

Objectives:

After completing this practical, students will be able to:

1. Design a basic dashboard layout using multiple worksheets.
2. Add filters and interactive controls.
3. Implement Highlight and Filter Actions.
4. Use URL Actions and Set Actions for interactivity.
5. Create dynamic dashboards using parameters.
6. Design dashboards for multiple devices.
7. Build a complete data story using the Tableau Story feature.

STEP 1: CONNECT THE EXCEL FILE

1. Open **Tableau Public/Desktop**.
2. Click **Microsoft Excel**.
3. Select the Excel file:

   Retail_Store_Performance_500 (1).xlsx

4. Drag the required Excel sheet to the Data Source area, if required.
5. Click **Sheet 1** to begin creating visualizations.

STEP 2: CHECK DATA TYPES

Before creating the worksheets, verify that the fields are correctly identified.

Dimensions:

- Region
- State
- City
- Customer Segment
- Gender
- Age Group
- Product Category
- Product Name
- Shipping Mode

- Payment Mode
- Returned

Measures:

- Quantity
- Unit Price
- Discount (%)
- Sales
- Profit
- Customer Rating
- Sales Target

Date Field:

- Order Date

If **Order Date** is not recognized as a date, click its data type icon and select **Date**.

PART A: CREATE WORKSHEETS

The following worksheets will be created:

1. Sales by Region
2. Sales by Category
3. Monthly Sales Trend
4. Profit by Region
5. Customer Rating Analysis

SHEET 1: SALES BY REGION

Step 1:-
Rename **Sheet 1** as: **Sales by Region**

Step 2:-
Drag **Region** to the **Rows** shelf.

Step 3:-
Drag **Sales** to the **Columns** shelf.

Step 4:-
Click **Show Me**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Drag **Sales** to **Label** on the Marks card.

Expected Output:-
The worksheet will display sales for each region.

Region	Sales
North	Sales Value
South	Sales Value
East	Sales Value
West	Sales Value

SHEET 2: SALES BY PRODUCT CATEGORY

Step 1:-
Click **New Worksheet**.

Step 2:-
Rename the worksheet: **Sales by Category**

Step 3:-
Drag **Product Category** to **Rows**.

Step 4:-
Drag **Sales** to **Columns**.

Step 5:-
Select **Bar Chart** from Show Me.

Step 6:-
Drag **Profit** to **Color** on the Marks card.

Step 7:-
Drag **Sales** to **Label**.

Expected Output:-
A bar chart will display sales for different product categories, with the bars colored according to Profit.

SHEET 3: MONTHLY SALES TREND

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Monthly Sales Trend**

Step 3:-
Drag **Order Date** to **Columns**.

Step 4:-
Click the drop-down arrow on **Order Date**.

Step 5:-
Select **Month**.

Step 6:-
Drag **Sales** to **Rows**.

Step 7:-
Select **Line Chart**.

Expected Output:-
The worksheet will display the monthly sales trend over time.

SHEET 4: PROFIT BY REGION

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Profit by Region**

Step 3:-
Drag **Region** to **Columns**.

Step 4:-
Drag **Profit** to **Rows**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Drag **Profit** to **Label**.

Expected Output:-
A bar chart will display the profit generated by each region.

SHEET 5: CUSTOMER RATING ANALYSIS

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Customer Rating Analysis**

Step 3:-
Drag **Product Category** to **Columns**.

Step 4:-
Drag **Customer Rating** to **Rows**.

Step 5:-
Change the aggregation from **SUM** to **Average**.
The field should now display:
AVG (Customer Rating)

Step 6:-
Select **Bar Chart**.

Expected Output:-
A bar chart will display the average customer rating for each product category.

PART A: DESIGN A BASIC DASHBOARD

STEP 3: CREATE THE DASHBOARD

Step 1:-
Click the **New Dashboard** button at the bottom of Tableau.

Step 2:-
Rename the dashboard: **Retail Store Performance Dashboard**

Step 3:-
Set the dashboard size to:
Automatic
Or
Desktop Browser

STEP 4: ARRANGE THE WORKSHEETS

From the left-side dashboard pane, drag the required worksheets into the dashboard.

Suggested Dashboard Layout:-

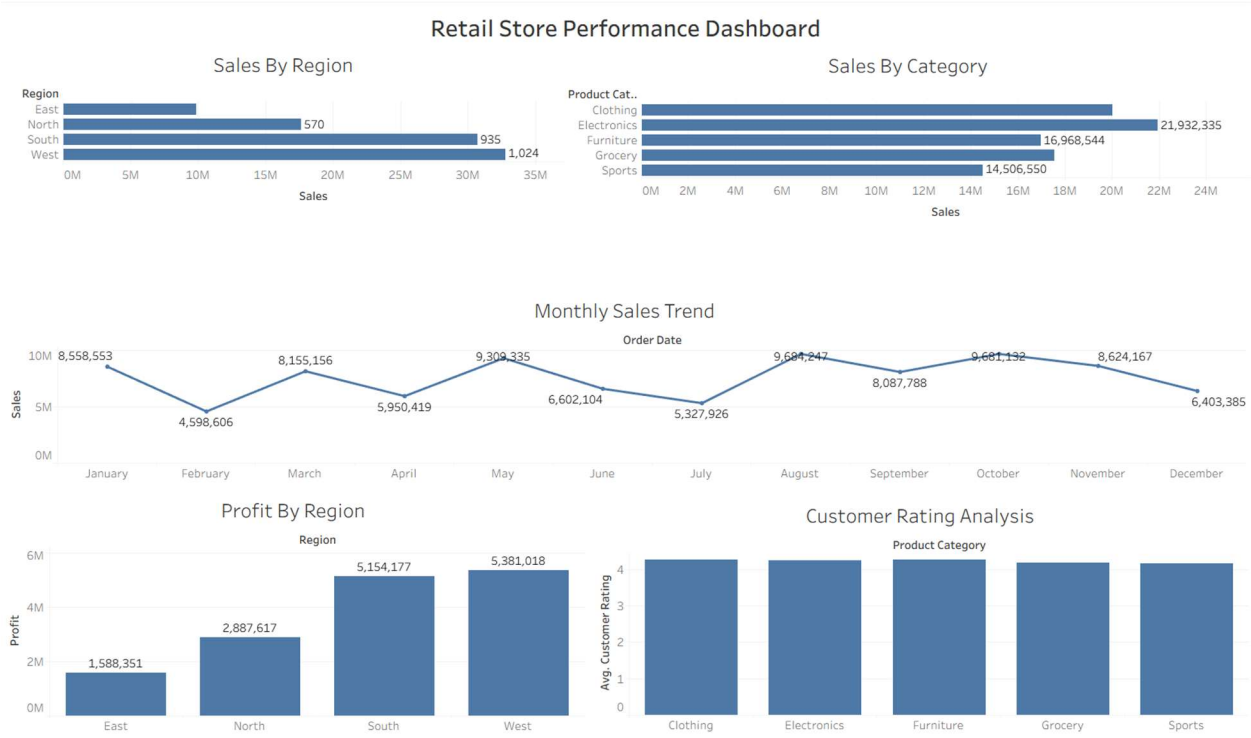
RETAIL STORE PERFORMANCE DASHBOARD

Region Filter Product Category Filter

Sales by Region	**Sales by Category**
Regional Sales	Category Sales
Monthly Sales Trend	
Monthly Sales Analysis	
Profit by Region	**Customer Rating Analysis**
Regional Profit	Customer Ratings

Use **Horizontal Containers** and **Vertical Containers** where required to organize the dashboard properly.

Expected Output:-



PART B: ADD FILTERS AND INTERACTIVE CONTROLS

STEP 5: ADD REGION FILTER

Step 1:-
Open any worksheet, such as **Sales by Region**.

Step 2:-
Drag **Region** to the **Filters** shelf.

Step 3:-
Keep all values selected.

Step 4:-
Right-click **Region** on the Filters shelf.

Step 5:-
Select: **Show Filter**
The Region filter will appear on the worksheet.

STEP 6: APPLY THE FILTER TO ALL WORKSHEETS

On the Region filter card:

1. Click the small drop-down arrow.
2. Select **Apply to Worksheets**.
3. Select **All Using This Data Source**.

Now the Region filter will control all worksheets that use the same data source.

Example

If **North** is selected, all charts in the dashboard will display data related to the North region.

STEP 7: ADD PRODUCT CATEGORY FILTER

Repeat the same process for Product Category.

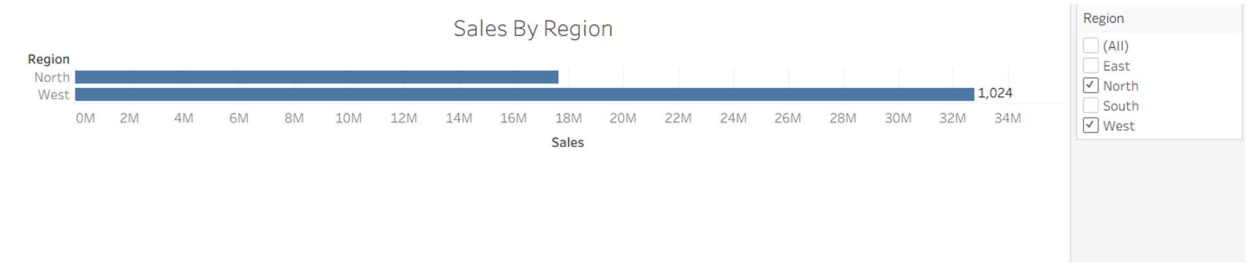
1. Drag **Product Category** to **Filters**.
2. Select **Show Filter**.
3. On the filter card, select:

Apply to Worksheets → All Using This Data Source

The dashboard will now contain two interactive filters:

- Region
- Product Category

Expected Output:-



PART C: HIGHLIGHT ACTION

A Highlight Action highlights related data without completely removing the other data from the visualization.

STEP 8: CREATE HIGHLIGHT ACTION

Step 1:-

Go to the dashboard.

Step 2:-

Select: **Dashboard → Actions**

Step 3:-

Click: **Add Action → Highlight**

Step 4:-

Configure the action as follows:

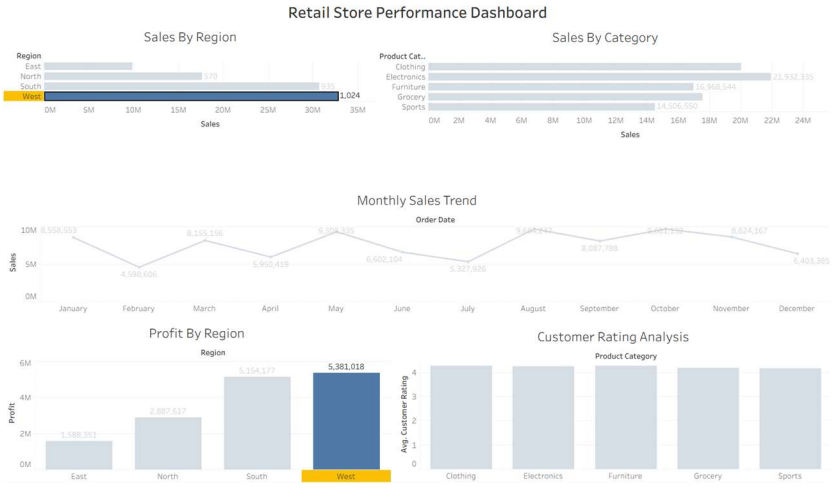
Option	Selection
Source Sheet	Sales by Region
Target Sheets	All other sheets
Run action on	Select
Clearing Selection	Show All Values

Step 5:-

Click **OK**.

Result:-

When a region such as **West** is selected in the Sales by Region chart, related information in the other charts will be highlighted.



PART C: FILTER ACTION

A Filter Action filters the data displayed in other worksheets based on the user's selection.

STEP 9: CREATE FILTER ACTION

Step 1:-

Go to: **Dashboard → Actions**

Step 2:-

Click: **Add Action → Filter**

Step 3:-

Configure the action as follows:

Option	Selection
Source Sheet	Sales by Region
Target Sheets	Sales by Category, Monthly Sales Trend, Profit by Region, Customer Rating Analysis
Run action on	Select
Clearing Selection	Show All Values

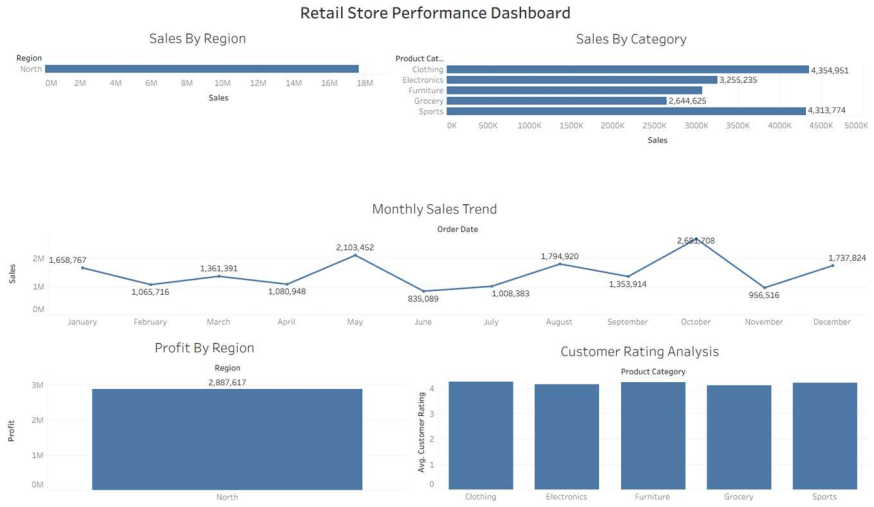
Step 4:-

Click **OK**.

Result:-

If **North** is selected in the Sales by Region chart, the other charts will display only North Region data.

To return to the complete dataset, click in an empty area of the visualization.



PART D: URL ACTION

A URL Action can be used to open an external webpage when a user selects data in a dashboard.

STEP 10: CREATE URL ACTION

Step 1:-

Go to: **Dashboard → Actions**

Step 2:-

Select: **Add Action → Go to URL**

Step 3:-

Enter the action name: **Open Retail Website**

Step 4:-

Select the source sheet: **Sales by Category**

Step 5:-

Enter the website URL: **https://www.tableau.com**

Step 6:-

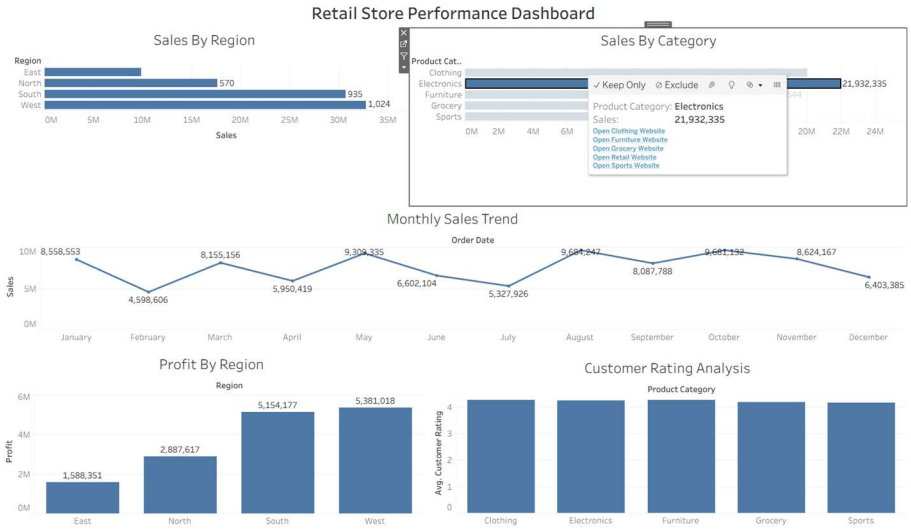
Set: **Run action on → Select**

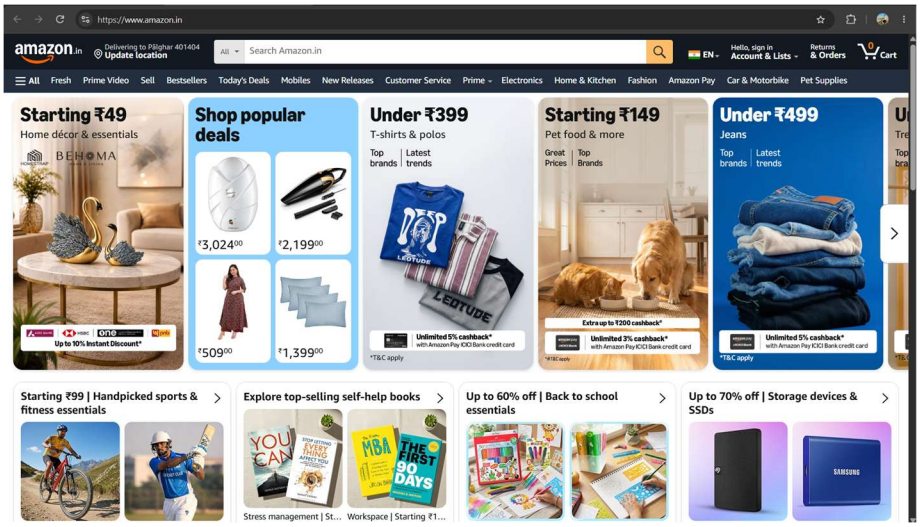
Step 7:-

Click **OK**.

Result:-

When a data item is selected in the selected worksheet, Tableau can open the specified webpage.





PART D: SET ACTION

A Set Action allows users to dynamically change the values contained in a Tableau set. In this practical, a set will be created for **Product Category**.

STEP 11: CREATE PRODUCT CATEGORY SET

Step 1:-

Open the **Sales by Category** worksheet.

Step 2:-

Right-click **Product Category**.

Step 3:-

Select: **Create → Set**

Step 4:-

Enter the set name: **Selected Product Category**

Step 5:-

Select all values.

Step 6:-

Click **OK**.

STEP 12: USE THE SET FOR COLOR

Step 1:-
Drag **Selected Product Category** to **Color** on the Marks card.
Selected and non-selected categories will now have different appearances.

STEP 13: CREATE SET ACTION

Step 1:-
Return to the dashboard.

Step 2:-
Go to:
Dashboard → Actions

Step 3:-
Select: **Add Action → Change Set Values**

Step 4:-
Configure the action:

Option	Value
Source Sheet	Sales by Category
Target Set	Selected Product Category
Run action on	Select
Clearing Selection	Keep Set Values

Step 5:-
Click **OK**.

Result:-
When a product category is selected, the **Selected Product Category** set will change dynamically.

PART E: CREATE A DYNAMIC DASHBOARD USING A PARAMETER

The parameter will allow the user to switch between:

- Sales
- Profit
- Sales Target

STEP 14: CREATE A PARAMETER

Step 1:-

In the Data Pane, right-click in an empty area.

Step 2:-

Select: **Create Parameter**

Step 3:-

Enter the following settings:

Setting	Value
Name	Select Measure
Data Type	String
Allowable Values	List

Step 4:-

Add the following values to the list:

- Sales
- Profit
- Sales Target

Step 5:-

Click **OK**.

Step 6:-

Right-click **Select Measure** and choose:
Show Parameter
The parameter control will now be visible.

STEP 15: CREATE A CALCULATED FIELD

Step 1:-
Right-click in the Data Pane.

Step 2:-
Select: **Create Calculated Field**

Step 3:-
Name the calculated field: **Dynamic Measure**

Step 4:-
Enter the following formula:

```
CASE [Select Measure]

WHEN "Sales" THEN [Sales]

WHEN "Profit" THEN [Profit]

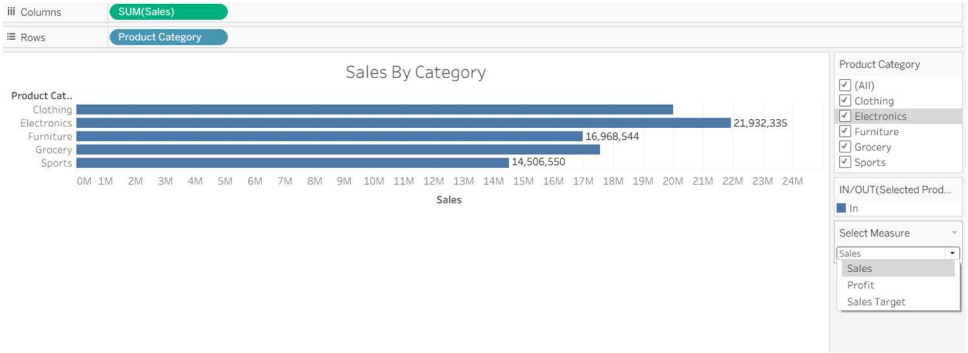
WHEN "Sales Target" THEN [Sales Target]

END
```

Step 5:-
Click **OK**.

TEST THE PARAMETER

Select **Sales**.
The chart will display Sales values.
Select **Profit**.
The chart will automatically change to Profit values.
Select **Sales Target**.
The chart will display Sales Target values.
This demonstrates how a parameter can dynamically change the measure used in a visualization.



STEP 16: CREATE A DYNAMIC CHART

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Dynamic Performance Analysis**

Step 3:-
Drag **Region** to **Columns**.

Step 4:-
Drag **Dynamic Measure** to **Rows**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Keep the **Select Measure** parameter visible.
The user can now switch between:
Sales → Profit → Sales Target
The chart will automatically update according to the selected measure.

Step 7:-
Add the **Dynamic Performance Analysis** worksheet to the dashboard.

PART F: DESIGN DASHBOARDS FOR MULTIPLE DEVICES
Tableau allows dashboards to be designed for different screen sizes and devices.
The dashboard can be customized for:

- Desktop
- Tablet
- Phone

STEP 17: CREATE A PHONE LAYOUT

Step 1:-
Open the **Retail Store Performance Dashboard**.

Step 2:-
Open **Device Preview** or **Device Designer**, depending on the Tableau version.

Step 3:-
Select: **Add Phone Layout**

Step 4:-
Arrange the charts vertically.

Suggested Phone Layout
RETAIL STORE PERFORMANCE
Region Filter
Product Category Filter
Sales by Region
Sales by Category
Monthly Sales Trend
Profit by Region
The charts should be arranged vertically so that they are easy to view on a mobile screen.

Other Device Layouts
Similarly, create layouts for:

- Desktop
- Tablet
- Phone

This ensures that the dashboard remains readable and interactive on different devices.



PART G: CREATE A COMPLETE DATA STORY

Tableau's **Story** feature can be used to present the analysis step by step.
The following story points will be created:

1. Overall Retail Performance
2. Regional Sales Analysis
3. Product Category Analysis
4. Monthly Sales Trend
5. Profit Analysis
6. Customer Rating Analysis
7. Dynamic Performance Analysis
8. Final Insights and Conclusion

STEP 18: CREATE A NEW STORY

Step 1:-

At the bottom of Tableau, click the **New Story** icon.

Step 2:-

Rename the story: **Retail Store Performance Story**

Step 3:-

Set the story size to:
Automatic
Or
Desktop Browser

STORY POINT 1: OVERALL RETAIL PERFORMANCE

Step 1:-

Drag the complete **Retail Store Performance Dashboard** into the Story.

Step 2:-

Set the title: **Overall Retail Store Performance**

Caption

This dashboard provides an overall analysis of retail store sales, profit, product categories, customer ratings, and regional performance.

STORY POINT 2: REGIONAL SALES ANALYSIS

Step 1:-
Click **New Blank Point**.

Step 2:-
Drag **Sales by Region** into the Story.
Title: Regional Sales Performance

Caption:-
This analysis compares sales across different regions and helps identify the region with the strongest retail performance.

STORY POINT 3: PRODUCT CATEGORY ANALYSIS

Step 1:-
Click **New Blank Point**.

Step 2:-
Drag **Sales by Category** into the Story.
Title:- Product Category Performance

Caption:-
This analysis compares sales and profit across different product categories to identify high-performing categories.

STORY POINT 4: MONTHLY SALES TREND

Step 1:-
Create a new Story Point.

Step 2:-
Drag **Monthly Sales Trend** into the Story.
Title:- Monthly Sales Trend Analysis

Caption:-
This visualization shows how retail sales change over time and helps identify periods of higher and lower sales performance.

STORY POINT 5: PROFIT ANALYSIS

Step 1:-
Create another Story Point.

Step 2:-
Drag **Profit by Region** into the Story.
Title:- Regional Profit Analysis

Caption:-
This analysis compares profit generated by different regions and identifies the most profitable areas.

STORY POINT 6: CUSTOMER RATING ANALYSIS

Step 1:-
Create a new Story Point.

Step 2:-
Drag **Customer Rating Analysis** into the Story.
Title:- Customer Satisfaction Analysis

Caption:-
This analysis compares average customer ratings across product categories to understand customer satisfaction.

STORY POINT 7: DYNAMIC PERFORMANCE ANALYSIS

Step 1:-
Create another Story Point.

Step 2:-
Add the **Dynamic Performance Analysis** worksheet.
Title:- Dynamic Retail Performance

Step 3:-
Use the parameter to demonstrate switching between:

- Sales
- Profit
- Sales Target

Caption:-

The parameter allows users to dynamically change the measure and analyze different aspects of retail store performance.

FINAL STORY POINT: CONCLUSION

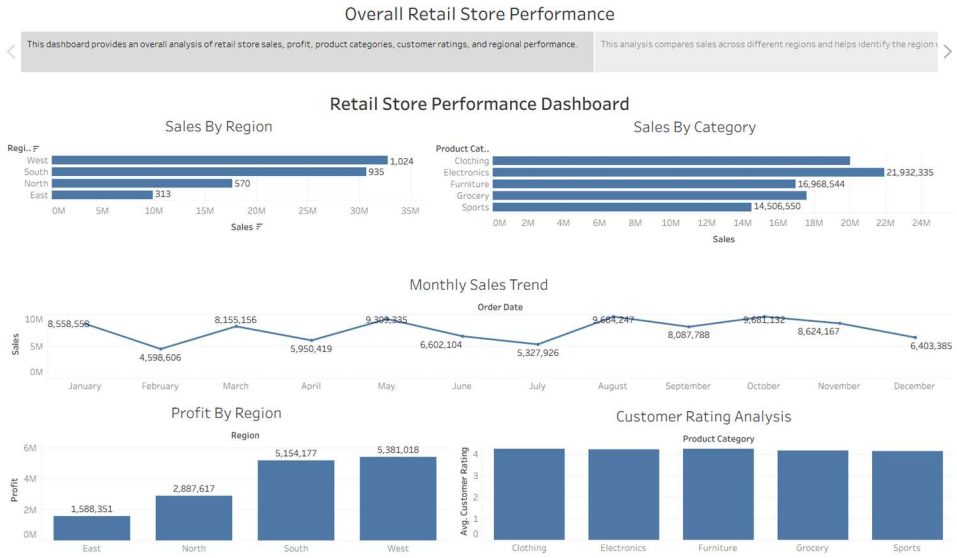
Step 1:-

Add the complete dashboard again or create a summary view.
Title:-Final Insights and Conclusion

Conclusion:-

The Retail Store Performance dashboard provides an interactive analysis of regional sales, product category performance, profit, customer ratings, and sales trends. Interactive filters, Highlight Actions, Filter Actions, Set Actions, URL Actions, and parameters allow users to explore the data dynamically. The Story feature presents the complete analysis step by step and helps users understand important business insights.

Excepted Output:-



OVERALL WORKFLOW:

Retail Store Performance Dataset
↓
Worksheets

1. Sales by Region
2. Sales by Category
3. Monthly Sales Trend
4. Profit by Region
5. Customer Rating Analysis
6. Dynamic Performance Analysis

↓
Interactive Dashboard

- Region Filter
- Product Category Filter
- Highlight Action
- Filter Action
- URL Action
- Set Action
- Parameter

↓
Device Layouts

- Desktop
- Tablet
- Phone

↓
Data Story

1. Overall Retail Performance
2. Regional Sales Analysis
3. Product Category Analysis
4. Monthly Sales Trend
5. Profit Analysis
6. Customer Rating Analysis
7. Dynamic Performance Analysis
8. Final Insights and Conclusion

Conclusion:-

This practical successfully demonstrated how to create an interactive retail store performance dashboard and data story using Tableau. Multiple worksheets were combined into a single dashboard, with filters and interactive controls added to make data analysis easier. Highlight, Filter, URL, and Set Actions were implemented to improve user interaction. Parameters were also used to create dynamic dashboards, while device-specific layouts ensured proper visualization on different screen sizes.